

Air University



Network Security (Lab)

Year 2026

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Lab Task # 02

Task 01: Linux Can Ping

First, ping the host, and then ping a website to verify that Linux can send requests.

```
(kali@kali)-[~]
$ ping 172.18.18.204
PING 172.18.18.204 (172.18.18.204) 56(84) bytes of data.
64 bytes from 172.18.18.204: icmp_seq=1 ttl=128 time=1.24 ms
64 bytes from 172.18.18.204: icmp_seq=2 ttl=128 time=0.902 ms
64 bytes from 172.18.18.204: icmp_seq=3 ttl=128 time=0.831 ms
64 bytes from 172.18.18.204: icmp_seq=4 ttl=128 time=0.764 ms
64 bytes from 172.18.18.204: icmp_seq=5 ttl=128 time=0.879 ms
^Z
zsh: suspended ping 172.18.18.204

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ ping youtube.com -c 4
PING youtube.com (142.250.202.238) 56(84) bytes of data.
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=1 ttl=128 time=47.4 ms
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=2 ttl=128 time=52.7 ms
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=3 ttl=128 time=46.9 ms
64 bytes from lcmcta-ad-in-f14.1e100.net (142.250.202.238): icmp_seq=4 ttl=128 time=48.4 ms

— youtube.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 46.910/48.850/52.695/2.284 ms
```

Task 02: No Request from Outside

Created a rule for incoming traffic that restricts every ping request.

```
(root@kali)-[~]
# iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```

Validating: validating the implemented rule to determine whether it was successful.

```
(root@kali)-[~]
# iptables --list
Chain INPUT (policy ACCEPT)
target prot opt source destination icmp echo-request
DROP icmp -- anywhere anywhere

Chain FORWARD (policy ACCEPT)
target prot opt source destination

Chain OUTPUT (policy ACCEPT)
target prot opt source destination
```

Verification: Ping the Linux from outside to see that the rule is successfully blocking the request.

```
C:\Users\HP>ping 192.168.180.129

Pinging 192.168.180.129 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.180.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 03: Linux Can Only Access Allowed Websites

The first two commands allow DNS (port 53) for domain name resolution. The next two commands permit HTTP/HTTPS traffic exclusively to 69.43.111.82, while the final two commands block all other outbound web traffic on ports 80 and 443.

```
(root@kali)-[~]
# sudo iptables -A OUTPUT -p udp --dport 53 -j ACCEPT

(root@kali)-[~]
# sudo iptables -A OUTPUT -p tcp --dport 53 -j ACCEPT

(root@kali)-[~]
# sudo iptables -A OUTPUT -p tcp -d 69.43.111.82 --dport 80 -j ACCEPT

(root@kali)-[~]
# sudo iptables -A OUTPUT -p tcp -d 69.43.111.82 --dport 443 -j ACCEPT

(root@kali)-[~]
# sudo iptables -A OUTPUT -p tcp --dport 80 -j REJECT

(root@kali)-[~]
# sudo iptables -A OUTPUT -p tcp --dport 443 -j REJECT

(root@kali)-[~]
# iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

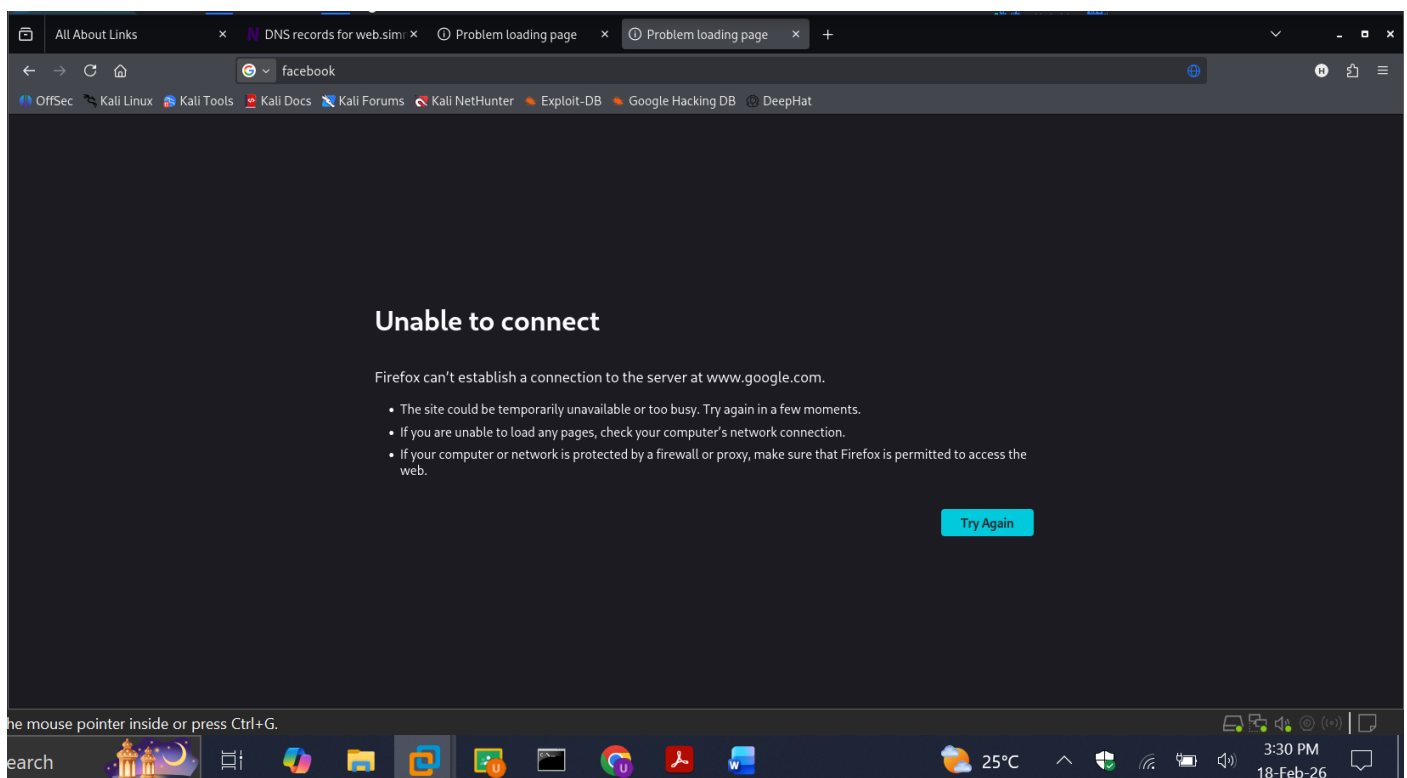
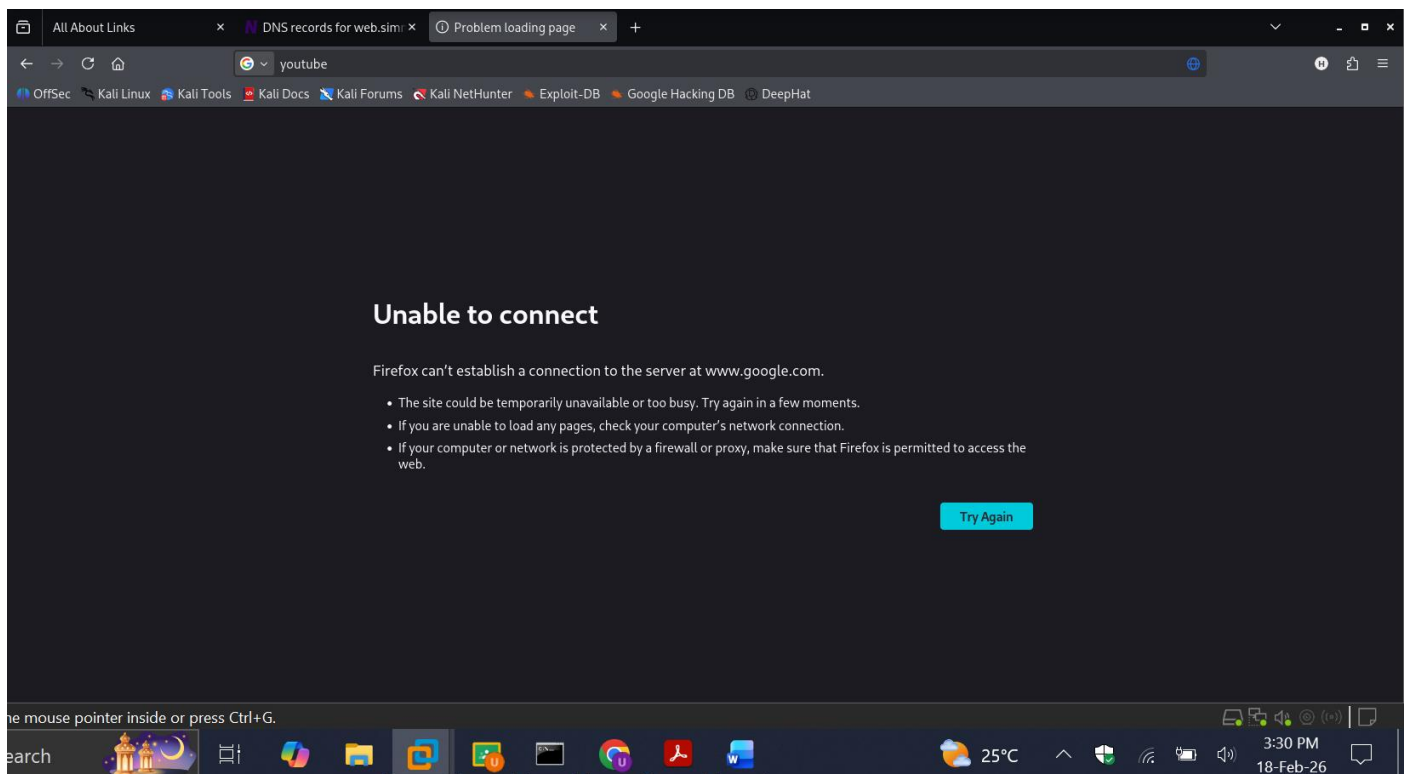
Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
ACCEPT     udp  --  anywhere              anywhere             udp dpt:domain
ACCEPT     tcp  --  anywhere              anywhere             tcp dpt:domain
ACCEPT     tcp  --  anywhere              web.simmons.edu      tcp dpt:http
ACCEPT     tcp  --  anywhere              web.simmons.edu      tcp dpt:https
REJECT     tcp  --  anywhere              anywhere             tcp dpt:http reject-with icmp-port-unreachable
REJECT     tcp  --  anywhere              anywhere             tcp dpt:https reject-with icmp-port-unreachable
```

Choose the Ip 69.43.111.82 for allowing and rejecting every other.

The screenshot shows a web browser window with the Nslookup.io website. The address bar shows the URL www.nslookup.io/domains/web.simmons.edu/dns-records/. The page title is "DNS records for web.simmons.edu.". Below the title, there are tabs for "Cloudflare", "Google DNS", "Authoritative", "Local DNS", "DMARC Checker", "SPF Checker", "DKIM Checker", and "Domain security check". The "Cloudflare" tab is selected. A message asks "Want to monitor this domain?" with a "Save / Track This Domain" button. Below this, a note states: "The Cloudflare DNS server responded with these DNS records. Cloudflare will serve these records for as long as the time to live (TTL) has not expired. After this period, Cloudflare will update its cache by querying one of the authoritative name servers." Under the heading "A records", there is a table with two columns: "IPv4 address" and "Revalidate in". The table contains one record: "69.43.111.82" with a revalidation time of "1h". Under the heading "AAAA records", there are no records listed. The browser's taskbar at the bottom shows various icons, including a file explorer, a terminal, and a web browser. The system clock in the bottom right corner shows "3:28 PM" and "18-Feb-26".

IPv4 address	Revalidate in
69.43.111.82	1h

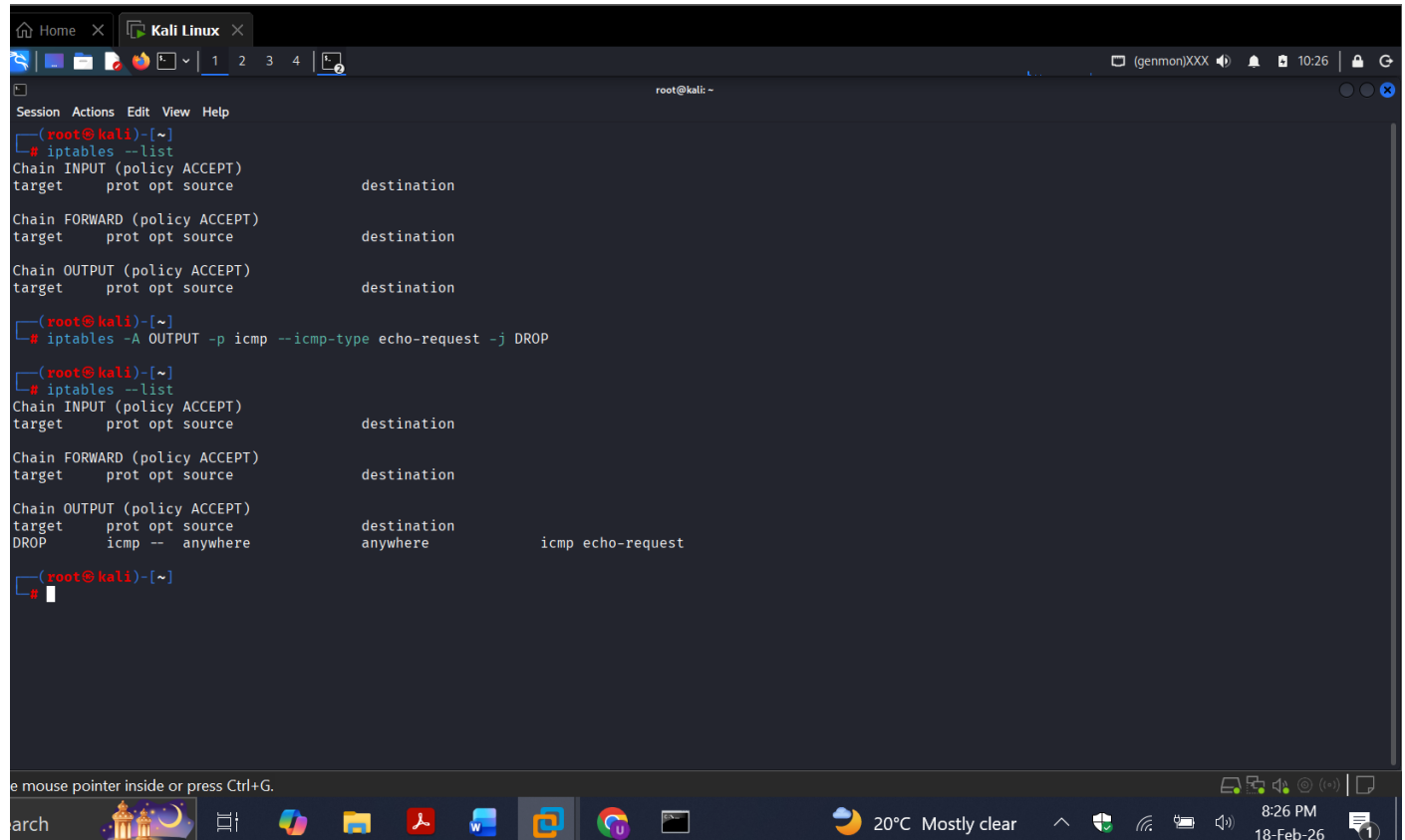
I verified by checking YouTube and Facebook to determine whether the rule has been applied.



Practice Tasks

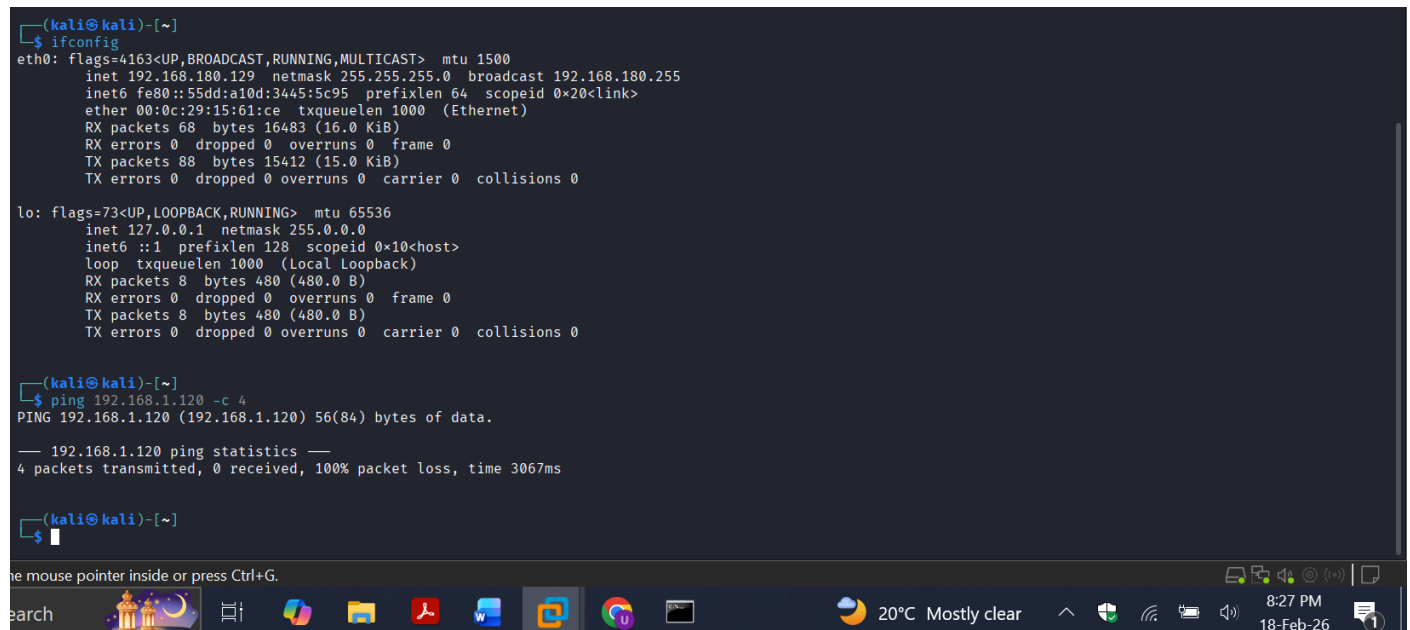
Create a rule to block outgoing ping packets:

Rule:



```
root@kali: ~  
Session Actions Edit View Help  
root@kali:~  
# iptables --list  
Chain INPUT (policy ACCEPT)  
target prot opt source destination  
  
Chain FORWARD (policy ACCEPT)  
target prot opt source destination  
  
Chain OUTPUT (policy ACCEPT)  
target prot opt source destination  
  
root@kali:~  
# iptables -A OUTPUT -p icmp --icmp-type echo-request -j DROP  
  
root@kali:~  
# iptables --list  
Chain INPUT (policy ACCEPT)  
target prot opt source destination  
  
Chain FORWARD (policy ACCEPT)  
target prot opt source destination  
  
Chain OUTPUT (policy ACCEPT)  
target prot opt source destination  
DROP icmp -- anywhere anywhere icmp echo-request  
  
root@kali:~  
#
```

Outgoing Request:



```
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.180.129 netmask 255.255.255.0 broadcast 192.168.180.255  
    inet6 fe80::55dd:a10d:3445:5c95 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:15:61:ce txqueuelen 1000 (Ethernet)  
    RX packets 68 bytes 16483 (16.0 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 88 bytes 15412 (15.0 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 8 bytes 480 (480.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 480 (480.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$ ping 192.168.1.120 -c 4  
PING 192.168.1.120 (192.168.1.120) 56(84) bytes of data.  
  
— 192.168.1.120 ping statistics —  
4 packets transmitted, 0 received, 100% packet loss, time 3067ms  
  
(kali@kali)-[~]  
$
```

Incoming Requests:

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : www.tendawifi.com
    Link-local IPv6 Address . . . . . : fe80::535b:5eee:ed52:f2d0%17
    IPv4 Address. . . . . : 192.168.1.120
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\HP>ping 192.168.180.129

Pinging 192.168.180.129 with 32 bytes of data:
Reply from 192.168.180.129: bytes=32 time<1ms TTL=64
Reply from 192.168.180.129: bytes=32 time<1ms TTL=64
Reply from 192.168.180.129: bytes=32 time<1ms TTL=64
Reply from 192.168.180.129: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.180.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\HP>
```

Block all incoming traffic:

Rule:

```
(root@kali)-[~]
# iptables -A INPUT -p tcp --dport 22 -j ACCEPT
(root@kali)-[~]
# sudo iptables -F INPUT DROP
(root@kali)-[~]
# iptables --list
Chain INPUT (policy DROP)
target prot opt source destination
ACCEPT tcp -- anywhere tcp dpt:ssh

Chain FORWARD (policy ACCEPT)
target prot opt source destination

Chain OUTPUT (policy ACCEPT)
target prot opt source destination

(root@kali)-[~]
#
```

Incoming traffic:

```
C:\Users\HP>ping 192.168.180.129

Pinging 192.168.180.129 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.180.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\HP>
```

Allow outgoing traffic:

Rule:

```
(root@kali)-[~]
# sudo iptables -P OUTPUT ACCEPT

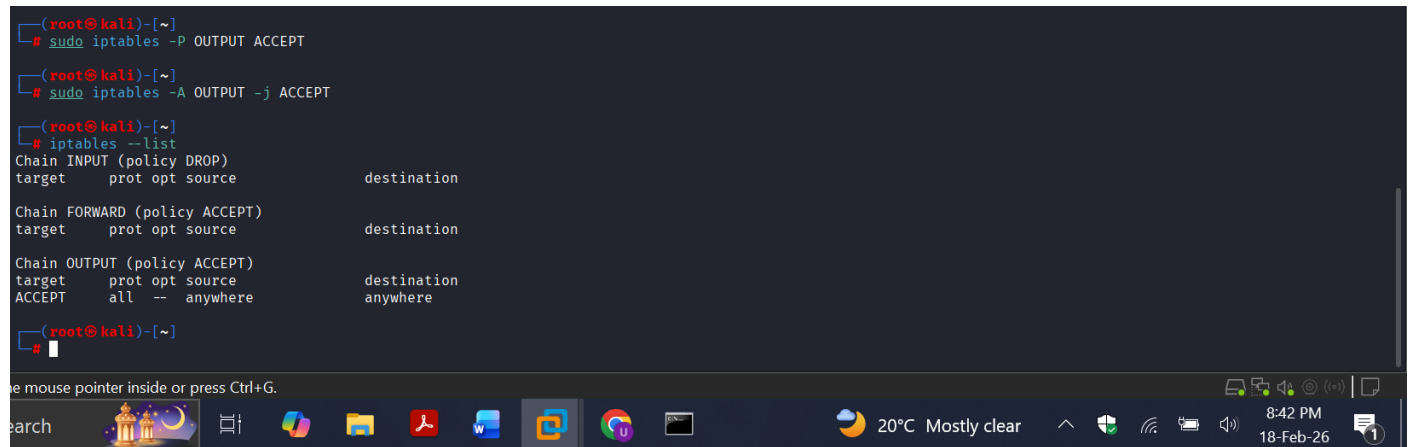
(root@kali)-[~]
# sudo iptables -A OUTPUT -j ACCEPT

(root@kali)-[~]
# iptables --list
Chain INPUT (policy DROP)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
ACCEPT    all  --  anywhere              anywhere

(root@kali)-[~]
#
```



Blocked incoming traffic:

Rule:

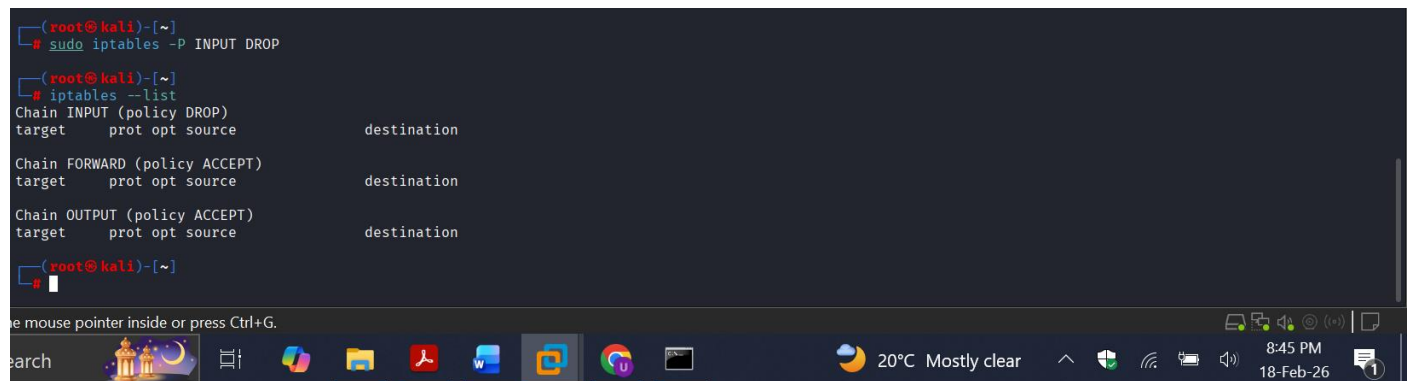
```
(root@kali)-[~]
# sudo iptables -P INPUT DROP

(root@kali)-[~]
# iptables --list
Chain INPUT (policy DROP)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

(root@kali)-[~]
#
```



THE END
